

Assignment: Day 9

Firewall Basics

Q1. Open Windows Defender Firewall settings on your computer and create a new outbound rule to block all traffic from the Spotify desktop app.

Ans:

Step	What to do
1	Press the Windows key, type Windows Defender Firewall with Advanced Security and open it (or press Win + R, type wf.msc and press Enter).
2	In the left pane click Outbound Rules , then click New Rule... in the right pane.
3	Select rule type Program and click Next.
4	Choose This program path and click Browse. Select the Spotify executable, usually C:\Users\<username>\AppData\Roaming\Spotify\Spotify.exe. Click Next.
5	Under Action, select Block the connection and click Next.
6	Keep Domain, Private and Public all ticked so the rule works on every network. Click Next.
7	Name the rule Block Spotify Outbound , add a short description, and click Finish .
8	To test, open Spotify. It cannot connect to the internet and shows an offline or no-connection message. The rule appears in the Outbound Rules list with a red block icon.

Setting	Value
Rule name	Block Spotify Outbound
Direction	Outbound
Program	Spotify.exe
Action	Block the connection
Profiles	Domain, Private, Public

Q2. Set up an inbound rule in Windows Defender Firewall to allow incoming connections only for Google Chrome on port 443 (HTTPS).

Ans:

Step	What to do
1	Open Windows Defender Firewall with Advanced Security and click Inbound Rules , then New Rule....
2	Select rule type Program and click Next.
3	Choose This program path and browse to Chrome: C:\Program Files\Google\Chrome\Application\chrome.exe. Click Next.
4	Under Action, select Allow the connection and click Next.
5	Tick Domain, Private and Public and click Next.
6	Name the rule Allow Chrome HTTPS 443 and click Finish .
7	The wizard does not ask for a port for a program rule, so double-click the new rule to open its Properties and go to the Protocols and Ports tab.
8	Set Protocol type to TCP , Local port to Specific Ports , and type 443 . Click Apply and then OK .

Setting	Value
Rule name	Allow Chrome HTTPS 443
Direction	Inbound
Program	chrome.exe
Action	Allow the connection
Protocol / Port	TCP / Local port 443
Profiles	Domain, Private, Public

Windows blocks unsolicited inbound connections by default, so with this rule Chrome is allowed to receive connections only on port 443.

Q3. Document, step-by-step, how you would block all traffic from a specific IP address (for example, 123.45.67.89) using Windows Defender Firewall.

Ans:

Step	What to do
1	Open Windows Defender Firewall with Advanced Security (Win + R, type wf . msc).
2	Click Inbound Rules , then New Rule...
3	Select rule type Custom and click Next.
4	Select All programs and click Next. Keep Protocol type: Any and click Next.
5	On the Scope page, go to the Remote IP addresses section and select These IP addresses . Click Add... , type 123 . 45 . 67 . 89, click OK, and then click Next. Leave the Local IP addresses set to Any.
6	Under Action, select Block the connection and click Next.
7	Tick Domain, Private and Public , click Next, name the rule Block IP 123.45.67.89 , and click Finish .
8	Repeat steps 2 to 7 under Outbound Rules so that traffic going to this IP address is blocked as well.

Setting	Value
Rule name	Block IP 123.45.67.89
Direction	Inbound and Outbound
Programs / Protocol	All programs / Any
Remote IP address	123.45.67.89
Action	Block the connection

Q4. Compare hardware firewalls (like those used in large offices) and software firewalls (like Windows Defender) by listing 3 key differences and a use case for each.

Ans:

Point	Hardware firewall	Software firewall
1. Location and protection	A separate physical device placed between the internet and the network. It protects every device behind it.	A program installed on one computer. It protects only that device.
2. Performance	Has its own processor and memory, so it handles heavy traffic without slowing down the computers.	Uses the computer's own CPU and RAM, so it can slow the device slightly.
3. Cost and management	Costly to buy and needs trained staff. Rules are managed centrally for the whole network and it offers advanced features such as VPN and intrusion prevention.	Cheap or free (built into Windows). Easy to set up, and rules are set separately on each device, including rules for individual apps.
Use case	A large office or company with hundreds of users, where the network needs one strong protection point at the entry.	A personal laptop or home PC, especially on public Wi-Fi, where the user wants to block or allow specific apps such as Spotify.